

Grades

 Print

Grade Item	Points	Weight Achieved	Grade	Comments and Assessments
Assignment 1	17.75 / 25	7.1 / 10	B	Overall Feedback total score: 17.75/29 You clearly put in a lot of effort and show strong intuition and engagement with the material, which is great to see. To improve your score, try to focus on precision: state the correct model, notation, and final result explicitly and concisely before adding explanations, as this will help your good understanding translate into full credit. ----- Deduction details: Q1(a) –0.5: You eventually reach the correct SSE/OLS objective, but the likelihood/log-likelihood derivation is confused (you mix up $p(x)$ vs $p(y x,w)$, mention integrating over x,w, and the steps are not a clean “log-likelihood → drop constants → SSE”). Q1(b) –0.5: You mention i.i.d. Gaussian noise, but you don’t clearly state the key assumption that X is fixed (or independent of the noise), and you don’t clearly state Bayesian LR assumptions beyond “posterior $\propto$ prior $\times$ likelihood” (missing an explicit prior statement / clarity). Q2(a) –0.25: Correct main conclusion that ΣN shrinks as $X^T X$ grows, but the explanation is a bit muddled (e.g., “non-proportional to N” phrasing) and doesn’t clearly mention shrinkage in informative eigen-directions. Q3(d) –0.5: You give generic covariance definitions, but you don’t explain the key point: it is the projection of parameter uncertainty through $\mathcal{X}^*$, so it depends on the direction/coverage of $\mathcal{X}^*$. Q3(e) –0.5: You describe MLE vs MAP generally, but you do not clearly make the required contrast: plug-in prediction treats $\mathcal{W}w$ as fixed and ignores the epistemic term (so variance doesn’t inflate in

poorly learned regions).

Q4(a) –0.25: Mostly right (narrow in data-rich, wide in data-poor), but you add some misleading “overfitting → variance to 0” language; the key point is widening away from observed regions.

Q4(b) –0.25: Partially right, but you don’t really connect to the **geometry**: $\mathbf{x}^\top \sum_N \mathbf{x}$ is small near well-spanned/covered directions and larger in poorly sampled directions.

Q5(a) –0.5: You propose “entropy-based uncertainty sampling,” but the expected simple answer is: choose $\mathbf{x}^*$ that **maximizes** $\beta^{-1} + \mathbf{x}^{*\top} \sum_N \mathbf{x}^*$ (or just the epistemic term).

Q5(b) –0.5: The justification should be “this targets directions that shrink Σ_N the most / poorly constrained regions,” but your answer focuses on decision-boundary/entropy ideas without tying to Bayesian LR informativeness.

Q6(a) –0.5: Plotting choices can be misleading (e.g., fixed data-space y-limits $[-1,1]$ may clip regression lines/data; $N=0$ “likelihood” handling is not cleanly justified).

Q6(b) –1.0: Multiple issues: (i) likelihood-shape description is inaccurate/over-specific and not tied to the constraint $w_0 + w_1 x_1 \approx y_1$; (ii) “posterior center” is estimated via grid argmax (MAP on a grid) instead of reporting the posterior mean m_N as requested; (iii) uncertainty labels/descriptions are ad-hoc (prior should be clearly broad at $N=0$).

Q7(a) –0.25: Good trends overall, but you incorrectly suggest precision growth looks the same “regardless of σ ”; in fact precision increases by $1/\sigma^2$ per sample, so larger $\sigma \Rightarrow$ slower precision growth.

Q7(b) –0.25: Correct direction (higher noise $\Rightarrow$ more cautious updates), but the link from σ to the update mechanics is not stated cleanly (learning-rate/precision dependence).

Q8(a) –0.5: You simulate drifting true temperature and generate new readings, but the Bayesian update uses the *old* measurement stream $\mathbf{x}_t$ rather than the newly generated drifting measurements, so it doesn’t match the intended experiment.

Q8(b) –0.5: Main conclusion is broadly right (static-parameter learner becomes overconfident and can’t track drift), but reasoning contains irrelevant claims and does not clearly state the needed fix (a dynamic/state-space model or forgetting/transition model).

Q9 –0.5: You end with the correct-looking form, but the derivation maximizes a single-sample likelihood rather than the joint likelihood for N i.i.d. samples, and includes incorrect/irrelevant claims (e.g., concavity/bell-curve intuition).

Q11(a) –1: Posterior is not derived correctly for N samples; the correct Gamma–Exponential conjugate update is not obtained.

Q11(b) –1: You conclude the prior is not conjugate, which is incorrect; the given prior is Gamma and is conjugate to the exponential likelihood.

-Q12 : 2/4

- **Q12(a) -0.5:** You did **not** simulate lifetimes from **Exponential(rate = $\theta_0 = 0.2$)** for $N=\{10,100\}$. Instead you generate **Uniform** samples and even set $\theta_0 = -0.2$ as an “initial weight”, which is not the required exponential data simulation.
- **Q12(b) -0.5:** You wrote the *exponential* MLE formula $\hat{\theta} = 1/\bar{x}$, but you compute it on **Uniform(0,1)** samples (and the printed “true theta” is inconsistent), so the reported MLE is **not valid for the required exponential data**.
- **Q12(c) -0.5:** You define the prior $p(\theta) = \theta^3 e^{-3\theta}$ correctly, but you build the **likelihood for a Uniform $U(0,\theta)$** model and then set `posterior = likelihood * prior` without deriving the required **Gamma posterior parameter update** for the exponential model.
- **Q12(d) -0.5:** Your conjugacy discussion is incorrect (you argue conjugacy fails based on “peaks/touching spikes” reasoning). For the intended model, the prior is **Gamma** and is **conjugate** to the exponential likelihood; your conclusion is not consistent with the theoretical conjugacy behavior.